IWMI Water Accounting (WA) Dashboard
A User Manual for Zambezi Basin

What is a WA dashboard?

The water accounting dashboard provides detailed multi-year water accounts of a basin's water resources including inflows, outflows, water use patterns and availability, to establish baseline conditions. It can also compare baseline and future water accounts (where available), to provide stakeholders with an understanding of the current and potential future water resources status.

The platform leverages advanced data visualization tools to offer user-friendly access to complex information, empowering stakeholders to understand the dynamics between various water-balance parameters. In addition, the dashboard was designed to ensure that it caters to stakeholders of diverse backgrounds and expertise, from policymakers to scientists. Stakeholder feedback was incorporated to enrich the platform's accessibility and user experience to enable active participation and engagement.

This web-based dashboard represents one of the tools that could be used in promoting effective stakeholder deliberation of water balance information for sustainable water resource management.

Why do we need a WA dashboard?

Water resource management is an essential global challenge, necessitating a comprehensive and inclusive approach to address the diverse needs of the stakeholders involved. Stakeholder deliberation is crucial in fostering cooperation and informed decision-making for sustainable water management. However, lack of hydrometeorological observations often limits our understanding of the complex interlinkages between competing demands for water in the basin or a country. To overcome the challenges of water data scarcity in managing water resources, the International Water Management Institute (IWMI) is generating reliable and systematic analysis ready water data products on water use, demand, water availability and scarcity using water accounting plus (WA+) framework. The water accounting plus (WA+) approach (Karimi et al., 2013) derives basin scale water availability and scarcity indicators using earth observation data products and limited in situ observations. The WA+ framework is an open-source python programming-based model that uses a collection of remote sensing data products and in situ data to quantify water accounting indicators such as a) water yield, b) irrigation/rainfed water use, c) productive/unproductive water use, and other water availability indicators at river basin scale. Continental water accounting plus (CWA+) is a modified version of WA+ where the water accounting indicators can be generated for any given boundary (catchment, or a country or a county).

The suite of water data generated by water accounting studies provides several new insights into water availability and scarcity indicators at continental scale. The availability of open-source codes enables repeatable and rapid water accounting assessments without much effort. IWMI's WA dashboard is a web-based dashboard built using Tableau technology to enhance stakeholder engagement and facilitate effective deliberation by summarizing and communicating the key water accounts basin.

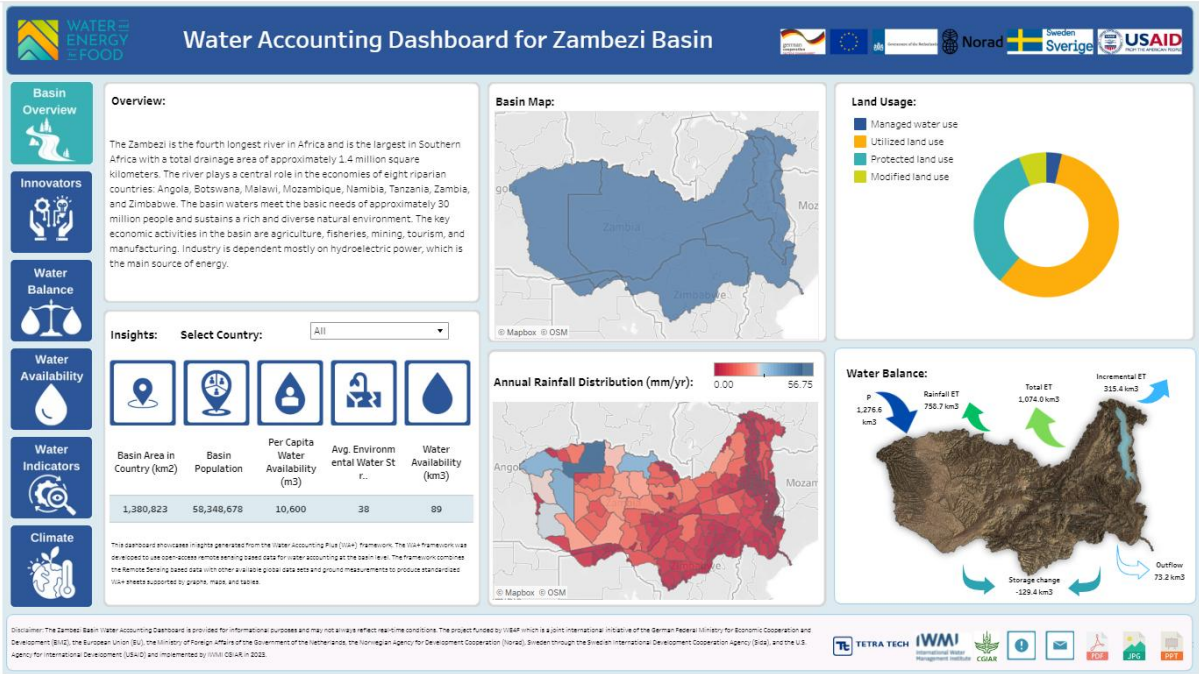

Figure 1. Illustration of the water accounting dashboard for the Zambezi Basin, Southern Africa.

Key Elements of the Zambezi Basin WA Dashboard

There are six key elements in the Zambezi Basin WA dashboard: i) basin overview ii) innovators iii) water balance iv) water availability v) water availability indicators and vi) climate. The description of each element and the contents of water accounts displayed under each element are presented in Figure 2 below.

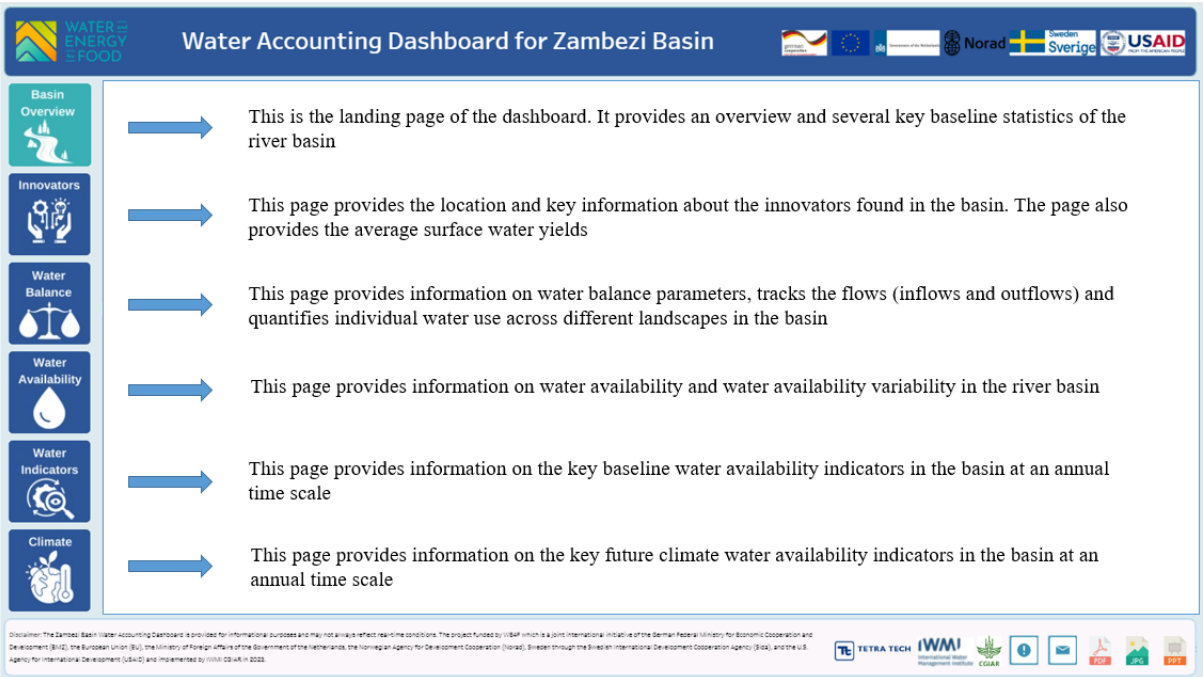

Figure 2. The key elements of Zambezi Basin water accounting dashboard.

1. Basin Overview

The landing page of the dashboard provides the basin overview information. Several key baseline statistics on the river basin are provided on this page. The description of each section is provided here.

Basin description: The basin overview page provides a brief synopsis of the dashboard. A basin description is provided to give a brief account of the basin hydrology and highlights important hydrologic challenges in the basin.

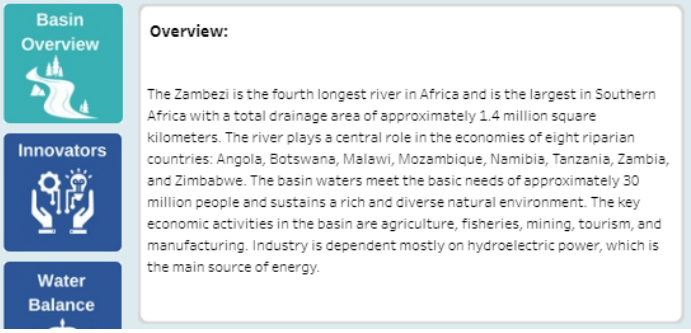

Figure 3. The description of the basin provided in the basin overview page

Annual Rainfall Distribution: The top-middle portion of the basin overview page provides a choropleth map of the average annual rainfall distribution. The rainfall is provided per district, and the map is interactive. The units used are millimeters per year.

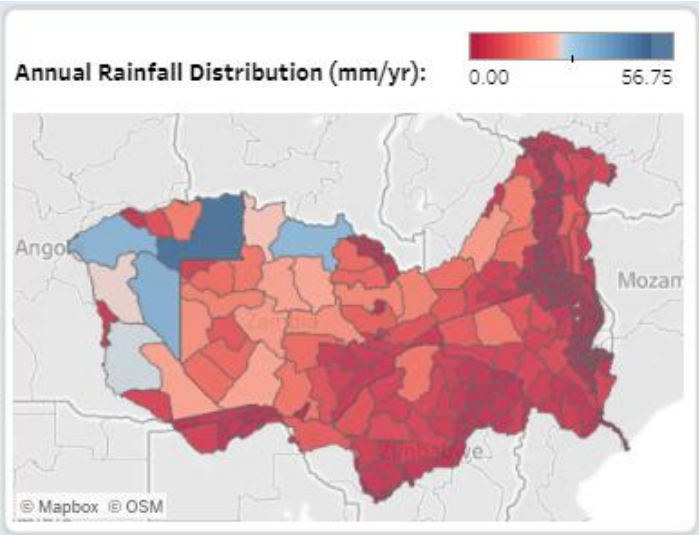

Figure 4. Basin average annual rainfall distribution

Land usage: At the top-right corner of the page is the distribution of land use classes across the basin. These are provided in a pie chart. The different land cover classes are reclassified into four broad classes of land use for water accounting (LUWA) classes. 1) The utilized land represents natural landscapes used in their natural forms, without modifying or altering water or land resources. For example, humans utilize forests, grasslands, and shrublands for grazing. 2) The Managed water class represents areas managed for agriculture where water is

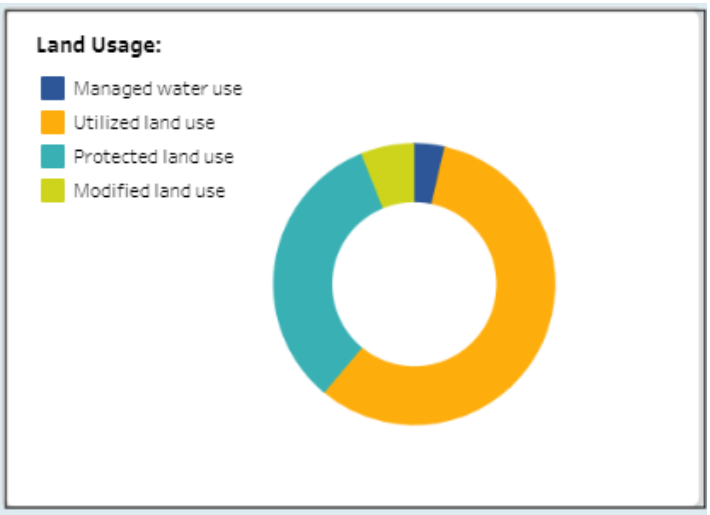

Figure 5. Land usage (km^2)

highly managed, such as irrigation. 3) Modified land represents an area where land is modified for human use. For example, the natural landscapes are cleared/modified to grow crops under rainfed conditions. 4) Protected land use defines the area classified as protected, such as national parks or other preserved areas.

Insights: The bottom-left portion of the basin overview page provides basic information on basin insights such as basin area, population, per-capita water availability, average environmental water stress and water availability for future use. These insights are interactive and provided for each country and the entire basin.

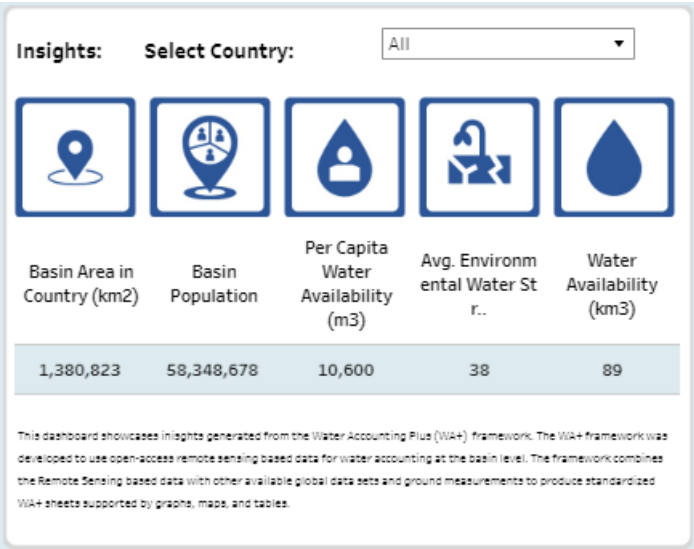

Figure 6. The basic information on basin insights

Basin Map: This is at the bottom of the page and is an interactive river basin map. Users can zoom in and out of the basin area overlaid on the world map using the + and – symbols. The home icon on the map will reset the map to the full extent of the basin. The triangle icon offers additional features for interacting with the map.

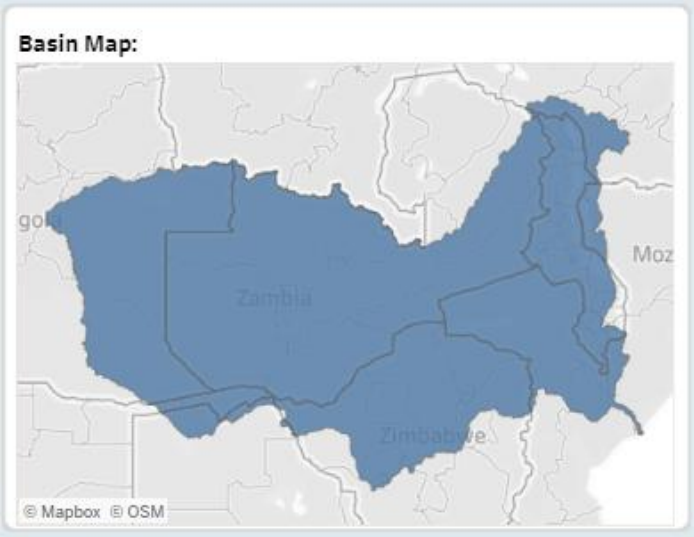

Figure 7. Basin map

Water Balance: The water balance figure shows modeled estimates of key water balance terms. Each figure is an average of the baseline period. The dark blue colored downward arrow represents the total volumetric annual precipitation (P, in km^3) received in the basin. The light green colored upward arrow represents total volumetric evapotranspiration (Total ET, in km^3). The lighter blue upward arrow represents Blue ET (Blue ET, in km^3), a portion of total ET, occurring from the blue water sources (surface water bodies, rivers, lakes or shallow groundwater aquifers). The small dark green upward arrow represents, Rainfall ET or Green ET (in km^3), a portion of total ET occurring from the green water sources (soil moisture replenished by the rainfall). The sum of Blue ET and Rainfall ET is equal to the total ET. The empty blue colored arrow represents basin outflow (in km^3) and the storage change denotes the changes in the basin storage due to either groundwater abstraction (+ve value) or groundwater recharge (-ve value).

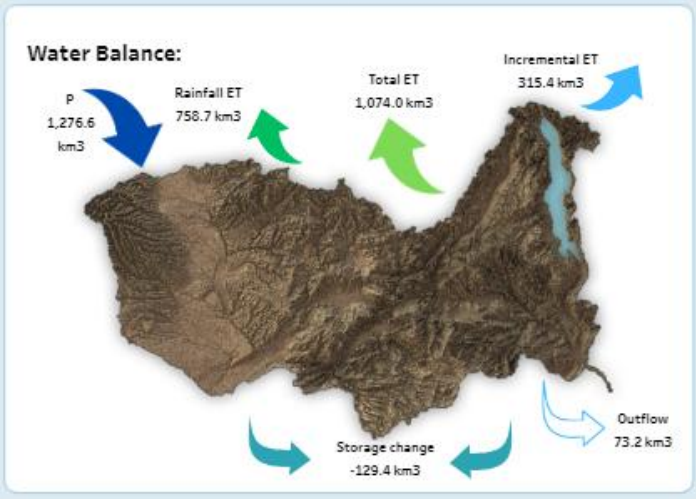

Figure 8. Basin water balance

2. Innovators

The Innovators page provides information on water availability for the innovators found within the basin. The initial overlay of the page provides an interactive map area which contains a district map of the average annual surface water yield per district. The page also comprises of a legend for the map, a brief description of surface water yield, a filter containing a list of innovation types, and two buttons labeled "Season 1 – Wet" and "Season 2 – Dry" (Figure 9). The list of innovation types includes the following:

(i) Battery stick kit/ULVA+ sprayer;
(ii) Biodigester;
(iii) Information services;
(iv) Logistics as a service;
(v) Marketplace;
(vi) Soil restoration for food, conservation, and markets;
(vii) Solar Dryer; and
(viii) Solar Water Pump.

The page initially provides the locations of different innovators and the innovations they have adopted within the basin. Hovering over each point provides information about the country, crop water, innovation type, end user, innovation, and innovator. Selecting a particular innovation type filters the map to display only what is of interest.

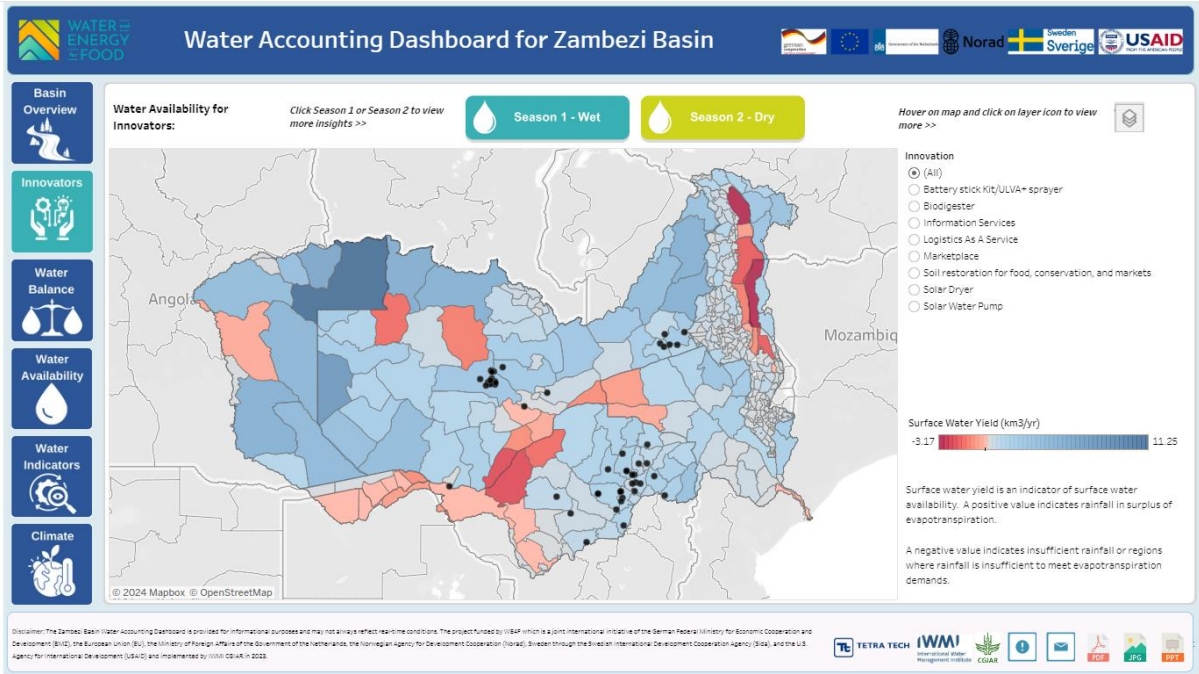

Figure 9. Initial overlay of the Innovators page

The Innovators page also provides the sustainability analysis results for the Zambezi Basin. This analysis focused on the cropped areas of the Zambezi Basin for both the dry and wet seasons and was carried out to assess the crop water availability for different crop water requirements. The crop water requirements assessed were 300-500mm, 400-600mm, 500-700mm, 700-900mm, 900-1100mm, and greater than 1100mm. The results of the sustainability analysis can be accessed by clicking on either the "Season 1 – Wet" or "Season 2 – Dry" buttons.

Selecting either the "Season 1 – Wet" or "Season 2 – Dry" buttons change the general overlay of the Innovators page (Figure 10). Six new buttons showing the different water requirements appear in between the "Season 1 – Wet" or "Season 2 – Dry" buttons. The map area changes to display the sustainability analysis results for the chosen crop water requirement. Each sustainability analysis map has three colors that represent the crop water availability for each crop water requirement. Green (SW) represents surface water, which implies precipitation and surface water are available enough to meet that crop water requirement for the selected season. Yellow (SWGW) implies there is a deficit in surface water availability. However, this deficit is less than the ground water availability and the crop water requirement can still be met. Pink (IW) represents areas in which the deficit is greater than the summation of SW and GW, and therefore, there is insufficient water to meet the selected crop water requirement for that season. On the left side of the page are a brief description of each level of crop water availability, a pie chart displaying the proportions of the different crop water availabilities, and a bar chart of the

areas of these in relation to the total cropped area and the total area. These statistics initially are representative of the entire basin.

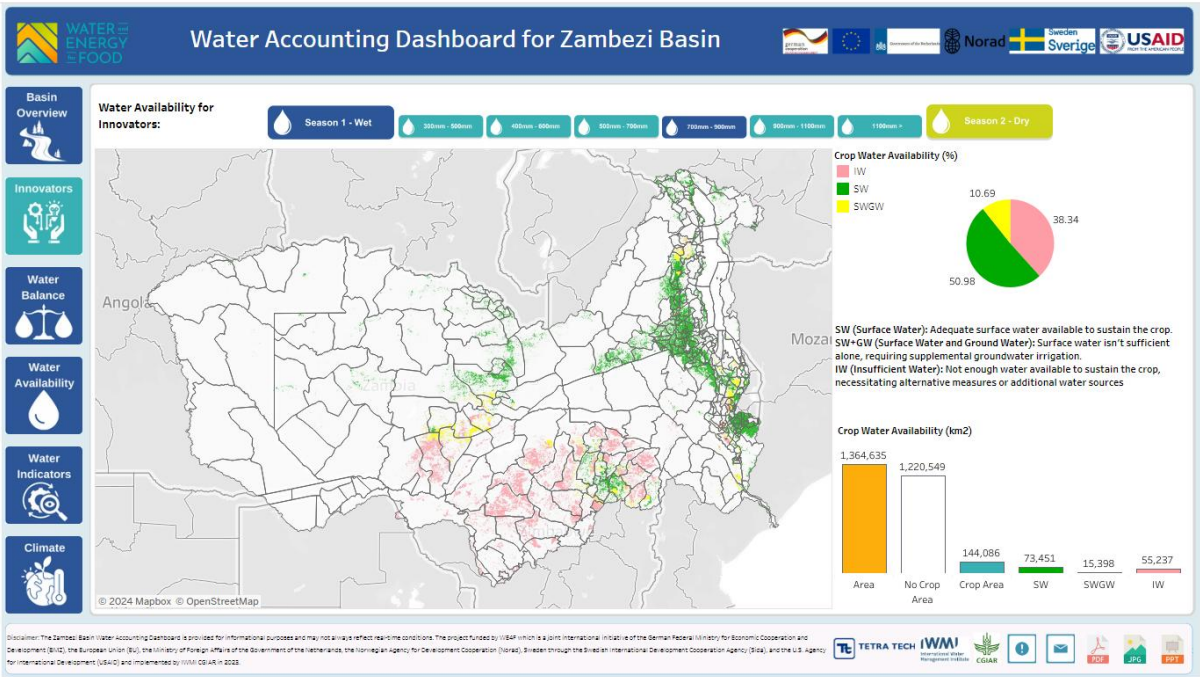

Figure 10: Sustainability analysis results of the Innovators page.

Selecting a district changes the pie chart and bar graph on the left to give statistics for that area. Information such as the district name, country, area, and percentages of each crop water availability are also given (Figure 11)

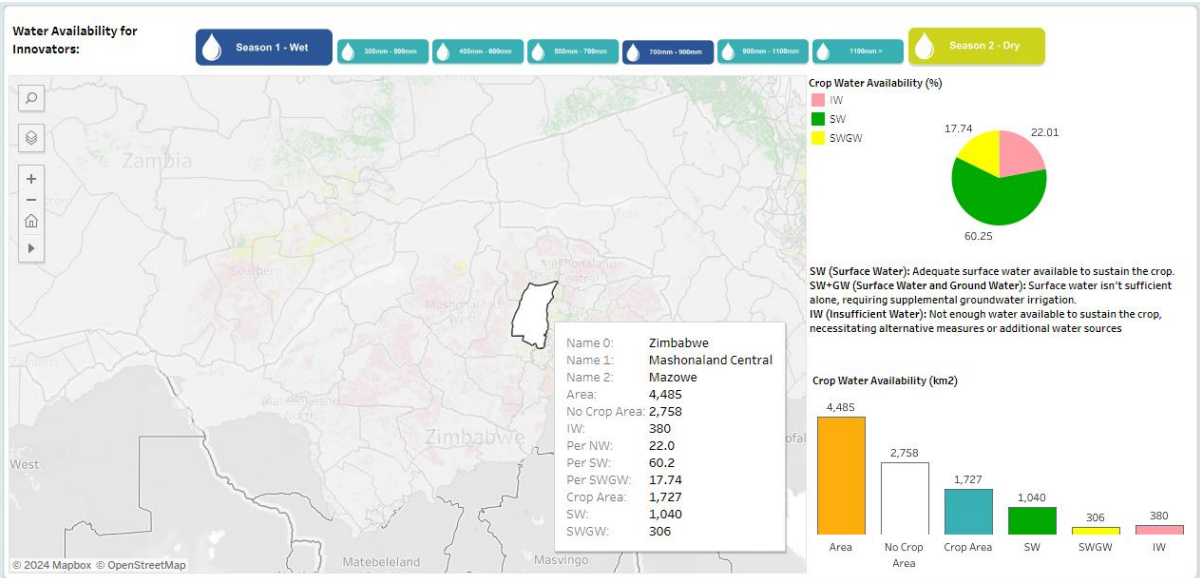

Figure 11: Selection of a districts on the Innovators page

Multiple districts can be selected at the same time and the statistics will change accordingly.

3. Water Balance

The Water balance, also known as the hydrologic balance or water budget, is a fundamental concept in hydrology. It refers to the equilibrium or accounting of water inputs, outputs, and

storage within a defined area, such as a watershed, catchment, or region. The water balance equation helps quantify the movement and distribution of water in various forms through the Earth's hydrological cycle.

Within the water accounting framework, a river basin's water balance is quantified and presented using several hydrologic variables. Unlike most hydrologic studies where the water balance is mostly represented by the key hydrologic variables such as precipitation, evapotranspiration, discharge and change in storage, the water accounting framework derives a variety of hydrologic parameters. A full list of water balance indicators quantified in the water accounting framework is presented in Table 1.

Table 1. List of hydrological variables and indicators quantified in the water accounting framework

No	Flux/Indicators	Description	Equation
1	*P* advection	Precipitation received in the basin, aggregated over the hydrologic year	$\sum_{i=1}^{12} P$
2	Basin inflow (interbasin transfer)	Surface water or groundwater diverted into the basin	Q_{in}^{sw} and Q_{in}^{gw} (Measured estimates)
3	Gross Inflow, *GI*	Total inflow from all sources	$P + Q_{in}^{sw} + Q_{in}^{gw}$
4	Change in the soil moisture, ΔSM	See equation 2.	See equation 2.
5	Net Inflow, *NI*	The gross inflow plus the change in soil moisture	$GI \pm \Delta SM$
6	ET rainfall, ET_{rain}	ETa that occurs from effective precipitation and canopy interception, summarized for all land cover classes (1 to n classes).	$\sum_{i=1}^{n} ET_{rain}$
7	ET incremental, ET_{incr}	ETa that occurs from other sources except effective precipitation and interception. Includes ET from irrigation water, groundwater abstraction, open water sources, summarized for all land cover classes (1 to n classes).	$\sum_{i=1}^{n} ET_{incr}$
8	Landscape ET, ETa_{land}	ETa from natural landscapes (protected, utilized and modified land use classes); not due to water management.	$ET_{rain} + ET_{incr}$
9	Consumed water, C_{water}	Total ETa that occurs from all landscapes over all months	$\sum_{i=1}^{12} ET_a$
10	Utilized flow, $Uzed_{flow}$	ETa from managed water use (irrigated crops, managed reservoirs).	ET_{incr} from the managed water use class
11	Exploitable water, EX_{water}	The exploitable water is the amount of water that can potentially be used within the basin	$NI - ET_{landscape}$
12	Available water, *AW*	The water that is left after meeting ET and reserve flow requirements	$GI - ETa_{land} - Reserve\ Flows$
13	Utilizable outflow, $Uzble_{flow}$	The water that can be reallocated for further uses after accounting for reserved flows and utilized flows.	$EX_{water} - ER_{flow} - Uzed_{flow}$

14	Qsw outlet	The river outflow at the outlet of the basin	Q_{outlet}^{sw}
15	Basin outflow (interbasin transfer)	Surface water or groundwater diverted to areas outside the basin	Q_{out}^{sw} and Q_{out}^{gw}
16	Non-consumed water, NC_{water}	Total outflow	$Q_{outlet}^{sw} + Q_{out}^{sw} + Q_{out}^{gw}$

The WA+ framework tracks both the flow and consumption (depletion) that occur within the basin as water moves from the inlet to the outlet of the basin. In this context, water resources that are depleted are rendered unavailable for further use. Figure 12 below is a Sunburst chart illustrates the basin's water balance by organizing data into hierarchical, circular layers that estimate how water is partitioned and consumed across different landscapes. The inner rings represent the foundational inputs and total water balance components, such as Precipitation and Gross Inflow, while the outer rings break down these totals into granular accounting categories like Landscape ET, Natural Blue ET, and Surface Water Outflow (Qsw). Depletion accounting is used within these segments to categorize land cover into four broad areas: Protected conservation zones, Utilized areas with limited human influence (forests and pastures), Modified rainfed agricultural zones, and Managed water use classes where water is purposefully withdrawn from surface or groundwater. This circular framework effectively derives flow accounting parameters including available water, managed use, and non-consumed water to show the part-to-whole relationship of the basin's resources.

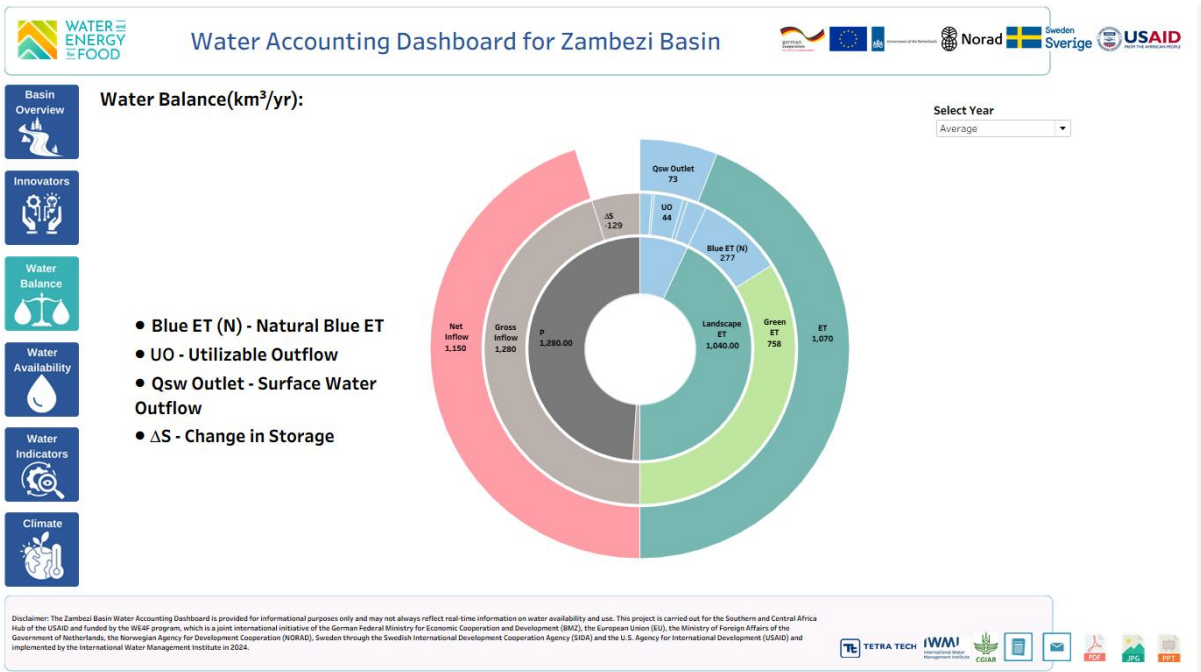

Figure 12. Basin water balance of the Zambezi Basin dashboard

4. Water Availability

Information on water availability in a river basin is crucial for understanding various aspects of human, environmental, and economic well-being. Through this dashboard, water availability in a river basin is summarized using indicators of water availability for a) humans b) environment c) agriculture and d) other uses.

The water availability page (Figure 13) provides an interactive district map and three line graphs that provide the trends of three parameters, namely rainfall, actual evapotranspiration (AET) and change in storage (Δs). The districts are coloured according to the country of location and selecting a single district on the map results in the provision of the three trends for that location. The three trends are summarised annually and are provided for the baseline (2003 to 2021) and climate periods (2027 to 2045).

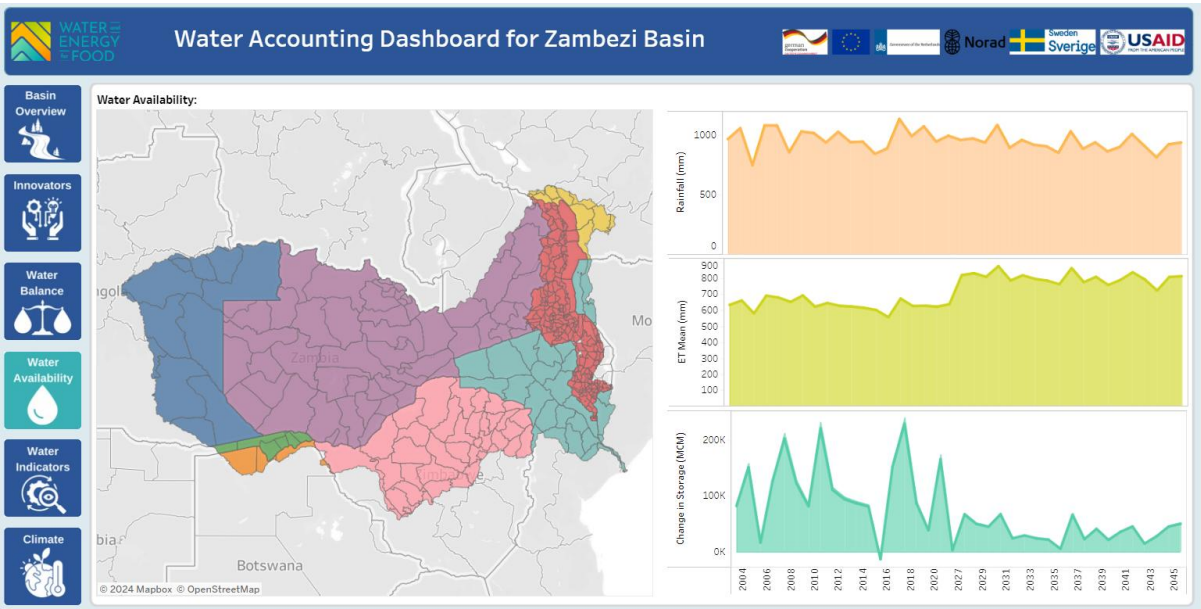

Figure 13. Water availability page of the Zambezi Basin dashboard

Rainfall is an inflow measured in millimetres and is provided as an annual average of the basin or a selected district. AET is an outflow also measured in millimetres and it is the actual water lost to the atmosphere from plant and other surfaces. Storage change provides information on water that is lost or gained to ground water. It is measured in million cubic metres (MCM). A negative value represents a loss of water to ground water, and hence, the amount of water that is recharged to aquifers. A positive value represents water that is abstracted from groundwater and supplied to the basin.

5. Water Availability Indicators

The key water availability indicators of the basin are summarized at an annual timescale. These indicators include rainfall ET, incremental ET, net inflow, and utilizable flow (Figure 14). Understanding inter-annual variability of water availability in the basin is crucial for developing sustainable water management plans.

Rainfall ET is the evapotranspiration of green water, and it is the fraction of actual total evapotranspiration due to precipitation. On the other hand, incremental ET is the evapotranspiration of blue water and is the fraction of total actual evapotranspiration that is not due to rainfall. In the WA framework, both rainfall and incremental ET are separated into the four broad classes of WA+ and these are protected land use, utilized land use, modified land use and managed water use. Rainfall ET and incremental ET sum up to depleted water. Net

inflow is the summation of gross inflow (precipitation plus any other inflows from surface or ground water that originate from outside the basin) and storage change. Utilizable outflow refers to water available for further water resources development after meeting all the basin demands of nature via landscape evapotranspiration, rainfed agriculture, domestic and industrial demand, and irrigated water use. This is the volume of water that can be used for planning any basin developmental activities such as additional diversion for domestic and industrial water use, additional irrigation development etc.

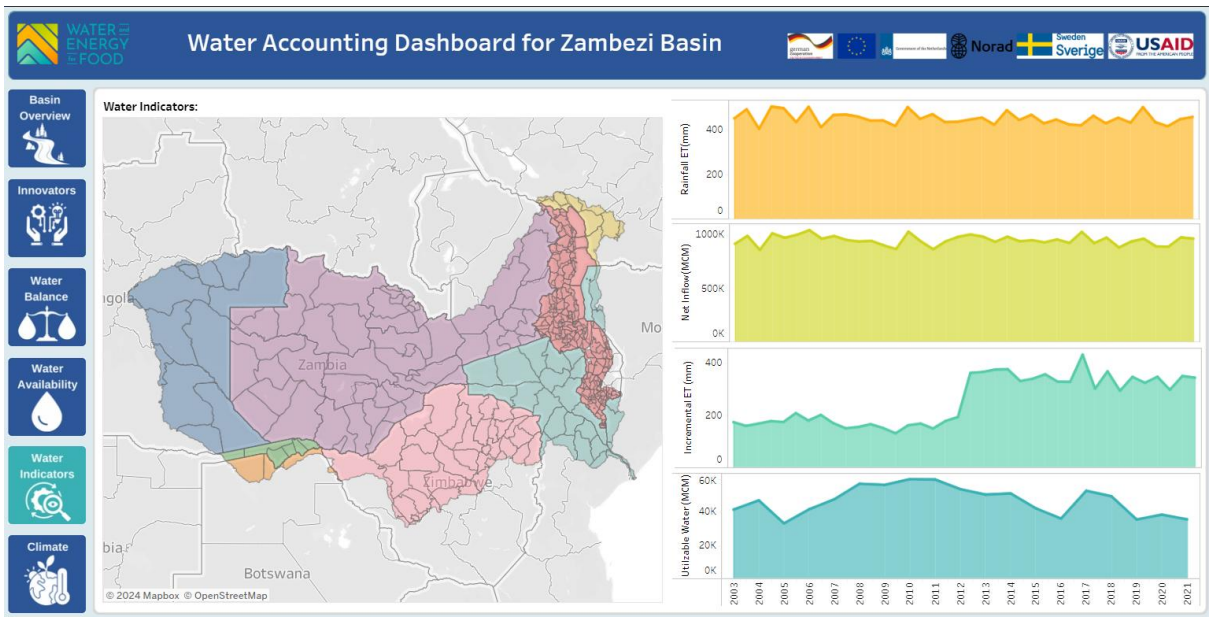

Figure 14. Water indicators page of the Zambezi Basin

6. Climate

The Climate page (Figure 15) provides information on future projections of water availability indicators. The Shared Socioeconomic Pathway which was used is SSP4.5 and it represents the middle way. This was conducted for three epochs which were the 2030s, 2035s and 2040s. Precipitation and maximum and minimum temperatures were obtained from four Global Circulation Models (GCMs), namely ASSESS-CM2, CNRM-CM6-1, MPI-ESM1-2LR and MIROC6. The relationships between precipitation, ET_a and ET_{ref} under current (baseline) conditions were established and assumed to hold for the future. Future ET_a and ET_{ref} were then estimated using the future precipitation and the minimum and maximum temperatures.

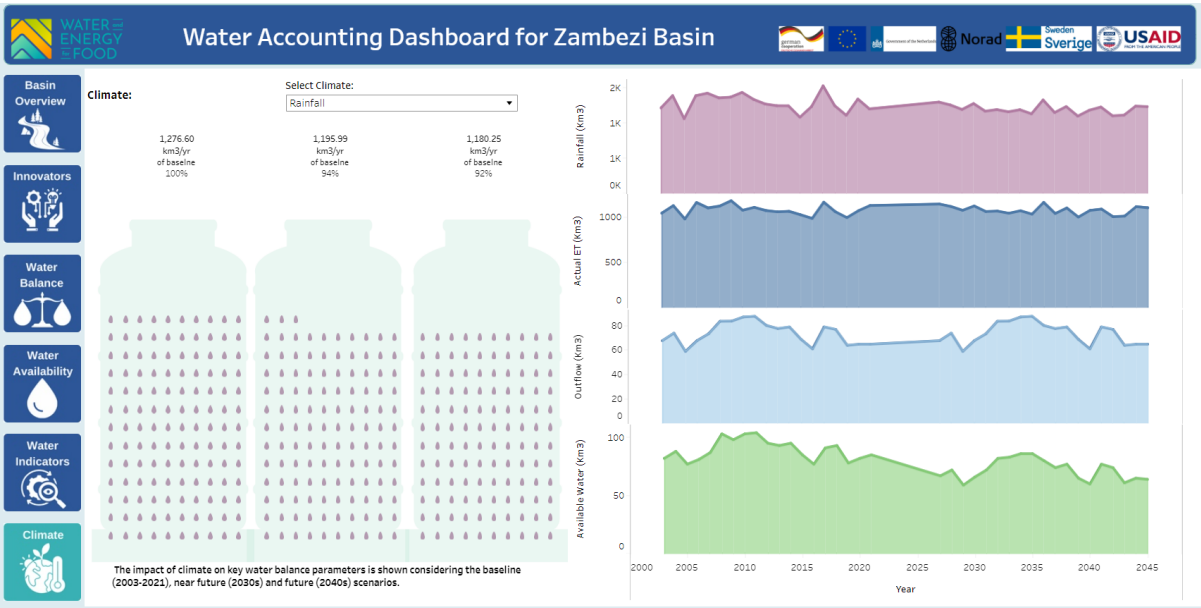

Figure 15: Climate page of the Zambezi Basin Dashboard

The page shows three bottles for the baseline, near future and future climate. The climate bottles display how the near future (2030s) and future (2040s) climate water volumes are proportional to that of the baseline volume. There are four indicators that can be selected, and these include rainfall, actual ET, outflow, and available water. All four indicators are in cubic kilometers and are also graphed on the right-hand side of the page.

The 2030s epoch shows the water volumes for the period 20272035, whereas the 2040s epoch shows the volumes for the period 20362045.

7. Additional features on the dashboard:

There are a few additional features available on the dashboard. A brief description and purpose of each icon on the lower right corner are presented here.

About: This icon provides general information about the project and the dashboard. The user manual and the technical manual can also be accessed by clicking this icon.

Contact: The contact icon provides email information on whom to contact in case you have any questions on the dashboard.

Printing options: The dashboard can be printed or saved using three options. The current view of the dashboard can be saved to the local computer in three different formats – a PDF document, JPG image or PowerPoint (PPT) presentation.